Homework 3

Due Date: October 27, 2025

1 - Hermite Interpolation

Given the function

$$f(x) = \frac{1}{1+25x^2}, \quad x \in [-1, 1],$$

with the function values and first derivatives at the nodes

$$x_0 = -1, \quad x_1 = 0, \quad x_2 = 1$$

Question 1.1

Construct the cubic Hermite interpolant $H(x)$ and evaluate $H(0.5)$.

Solution. The cubic Hermite interpolant on an interval $[x_0, x_1]$ is determined by the shape functions:

$$h_{00}(t) = 2t^3 - 3t^2 + 1, \quad h_{01}(t) = -2t^3 + 3t^2$$

$$h_{10}(t) = t^3 - 2t^2 + t, \quad h_{11}(t) = t^3 - t^2$$

Provided $h = x_1 - x_0$ and $t = \frac{x-x_0}{h} \in [0, 1]$. The cubic Hermite interpolant is given by:

$$H(x) = f(x_0)h_{00}(t) + f(x_1)h_{01}(t) + h\left[f'(x_0)h_{10}(t) + f'(x_1)h_{11}(t)\right]$$

We will construct the Hermite interpolant piecewise on the intervals $[-1, 0]$ and $[0, 1]$.

To start off with, the following table summarizes the function and derivative values at the nodes:

Node x_j	$f(x_j)$	$f'(x_j)$
-1	$\frac{1}{26}$	$\frac{50}{676} = \frac{25}{338}$
0	1	0
1	$\frac{1}{26}$	$-\frac{25}{338}$

Where $f'(x) = \frac{-50x}{(1+25x^2)^2}$.

On $[-1, 0]$, we have $h = 1$ and $t = x + 1$. So therefore we have:

$$h_{00}(x+1) = 2(x+1)^3 - 3(x+1)^2 + 1 = 2x^3 + 3x^2$$

$$h_{01}(x+1) = -2(x+1)^3 + 3(x+1)^2 = -2x^3 - 3x^2 + 1$$

$$h_{10}(x+1) = (x+1)^3 - 2(x+1)^2 + (x+1) = x^3 + x^2$$
$$h_{11}(x+1) = (x+1)^3 - (x+1)^2 = x^3 + 2x^2 + x$$

Substituting into the interpolant formula, we get:

$$H(x) = \frac{1}{26}(2x^3 + 3x^2) + 1(-2x^3 - 3x^2 + 1) + 1 \cdot \frac{25}{338}(x^3 + x^2) + 0$$

Simplifying, we have:

$$H(x) = -\frac{625}{338}x^3 - \frac{475}{169}x^2 + 1, \quad x \in [-1, 0]$$

On $[0, 1]$, we have $h = 1$ and $t = x$. So therefore our shape functions are as defined initially with x in place of t. Substituting into the interpolant formula, we get:

$$H(x) = 1(2x^3 - 3x^2 + 1) + \frac{1}{26}(-2x^3 + 3x^2) + 0 - \frac{25}{338}(x^3 - x^2)$$

Simplifying, we have:

$$H(x) = \frac{625}{338}x^3 - \frac{475}{169}x^2 + 1, \quad x \in [0, 1]$$

So the cubic Hermite interpolant is:

$$H(x) = \begin{cases} -\frac{625}{338}x^3 - \frac{475}{169}x^2 + 1, & x \in [-1, 0] \\ \frac{625}{338}x^3 - \frac{475}{169}x^2 + 1, & x \in [0, 1] \end{cases}$$

Evaluating at $x = 0.5$, we get that:

$$H(0.5) = \frac{625}{338}(0.5)^3 - \frac{475}{169}(0.5)^2 + 1 = \frac{1429}{2704} \approx 0.528479290$$

Question 1.2

Evaluate the absolution error of $f(0.5) - H(0.5)$, and show that the error agrees the prior estimate of

$$f(x) - p(x) = \frac{f^{(2n+2)}(\xi)}{(2n+2)!} \prod_{j=0}^{n} (x - x_j)^2$$

for some ξ between x and x_j.

Solution. Note that $f(0.5) = \frac{4}{29} \approx 0.1379310345$.

The absolute error with our estimate $H(0.5)$ is:

$$|f(0.5) - H(0.5)| = \left|\frac{4}{29} - \frac{1429}{2704}\right| = \frac{30625}{78416} \approx 0.390548$$

The fourth derivative of $f(x)$ is:

$$f^{(4)}(x) = \frac{15000\,(3125x^4 - 250x^2 + 1)}{(25x^2+1)^5}$$

The maximum of $|f^{(4)}(\xi)|$ for $\xi \in [-1, 1]$ occurs at $x = 0$, yielding:

$$|f^{(4)}(0)| = 15000$$

The product term evaluated at $x = 0.5$ is:

$$\prod_{j=1}^{2} (0.5 - x_j)^2 = (0.5-0)^2(0.5-1)^2 = \left(\frac{1}{2}\right)^2 \cdot \left(-\frac{1}{2}\right)^2 = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$

We also have that $(2n+2)! = 4! = 24$ for $n = 1$. So putting everything together, we can estimate the error as:

$$|f(0.5) - H(0.5)| \leq \frac{15000}{24} \cdot \frac{1}{16} = \frac{625}{16} \approx 39.0625$$

So therefore our absolute error is well within the established bound.

<u>Extra: Absolute Error with Standard Hermite Interpolant</u>

Note: Question 1.1 was originally done by hand with the regular Hermite Interpolant resulting in a polynomial of degree 5. This was the error analysis conducted for that version.

The absolute error with our estimate $P_5(0.5)$ is:

$$|f(0.5) - P_5(0.5)| = \left|\frac{4}{29} - \frac{6341}{10816}\right| = \frac{140625}{313664} \approx 0.4483300602$$

To estimate the error using the provided formula, we first need to compute the sixth derivative of $f(x)$, which works out to:

$$f^{(6)}(x) = \frac{11250000\,(109375x^6 - 21875x^4 + 525x^2 - 1)}{(25x^2+1)^7}$$

The maximum of $|f^{(6)}(\xi)|$ for $\xi \in [-1, 1]$ occurs at $x = 0$, yielding:

$$|f^{(6)}(0)| = 11250000$$

The product term evaluated at $x = 0.5$ is:

$$\prod_{j=0}^{2}(0.5 - x_j)^2 = (0.5+1)^2(0.5-0)^2(0.5-1)^2 = \left(\frac{3}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^2 \cdot \left(-\frac{1}{2}\right)^2 = \frac{9}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{9}{64}$$

We also have that $(2n+2)! = 6! = 720$ for $n = 2$.

So putting everything together, we can estimate the error as:

$$|f(0.5) - P_5(0.5)| \leq \frac{11250000}{720} \cdot \frac{9}{64} = \frac{140625}{64} \approx 2197.265625$$

So therefore our absolute error is well within the established bound.

Question 1.3

Obtain the cubic Hermite interpolant in Newton form $P(x)$., by constructing the Hermite divided-difference table.

Solution. The code for Lagrange Interpolation from the previous assignment was modified compute the Hermite Interpolation:

Listing 1: 1.3 Python

```python
import numpy as np

def f(x):
    return 1 / (1 + 25 * x**2)

def f_prime(x):
    return -50 * x / (1 + 25 * x**2)**2

def hermite_coefficients(nodes, f=f, f_prime=f_prime):
    # Parameterize nodes for Hermite interpolation
    z = np.concatenate((nodes, nodes))
    sorted_indexes = z.argsort()
    z = z[sorted_indexes]
    num_nodes = len(z)

    # Set up dd table with zeroth dd
    dd_table = np.array([[f(zi) for zi in z]])

    # First divided difference
    f_prime_nodes = np.array([f_prime(xi) for xi in nodes])
    zeros = np.zeros(len(nodes))
    first_dd = np.concatenate((f_prime_nodes, zeros))[sorted_indexes]
    for j in range(num_nodes - 1):
        if j % 2 == 1:
            first_dd[j] = (dd_table[0, j + 1] - dd_table[0, j]) / (z[j + 1] - z[j])
    dd_table = np.vstack([dd_table, first_dd])

    # Remaining Divided Differences
    for i in range(2, num_nodes):
        ith_dd = np.zeros(num_nodes)

        for j in range(num_nodes - i):
            # Calculate ith divided differences
            ith_dd[j] = (dd_table[i - 1, j + 1] - dd_table[i - 1, j]) / (z[j + i] - z[j])

        dd_table = np.vstack([dd_table, ith_dd])
```

```python

    coefficients = np.array([dd_table[i, 0] for i in range(dd_table.shape[0])])
    return coefficients, dd_table.T, z

def generate_hermite(nodes, n):
    # Get coefficients
    a = hermite_coefficients(nodes)[0]
    # Start with function and constant term
    equation = f"P_{{{2*n+1}}}(x) = {a[0]}"
    w = []
    # Build (x - xi) terms
    for xi in nodes:
        if abs(xi) <= 1e-14:
            w.append('(x)')
        elif xi < 0:
            w.append(f"(x + {abs(xi)})")
        else:
            w.append(f"(x - {xi})")
    b = ""

    # Build polynomial string
    for i in range(1, len(a)):
        for j in range(i):
            # Multiply (x - xi) terms
            b += '^2' if j%2 ==1 else w[int(j/2)]
        # Multiply (x - xi) product with current coefficient and add term
        if a[i] > 0:
            equation += f" + {a[i]}{b}"
        elif a[i] < 0:
            equation += f" - {abs(a[i])}{b}"
        b = ""

    return equation

if __name__ == '__main__':
    nodes_1 = np.array([-1,0])
    equation_1 = generate_hermite(nodes_1, 1)
    table_1, z_1 = hermite_coefficients(nodes_1)[1:3]

    nodes_2 = np.array([0,1])
    equation_2 = generate_hermite(nodes_1, 1)
    table_2, z_2 = hermite_coefficients(nodes_1)[1:3]

    equations = [equation_1,equation_2]
    tables = [table_1, table_2]
    z = [z_1,z_2]

```

```python
    with open("./outputs_3/hermite.txt", 'w') as file:
        for k in range(2):
            file.write(equations[k] +'\n\n')

            file.write('\\begin{center}\n')
            file.write('\\begin{tabular}{|c|c|c|c|c|c|}\n')
            file.write('\\hline\n')
            file.write('& $z_i$ & $f[z_i]$ & 1st dd. & 2nd dd. & 3rd dd. \\\\\n')
            file.write('\\hline\n')
            for i in range(tables[k].shape[0]):
                file.write(f'$z_{i}$ & ${z[k][i]}$ ')
                for j in range(tables[k].shape[1]):
                    file.write(f'& ${tables[k][i,j]:.4f}$ ')
                file.write('\\\\ \n')
            file.write('\\hline\n')
            file.write('\\end{tabular}\n')
            file.write('\\end{center}\n\n')
```

The divided-difference tables and Hermite Interpolants generated were:

On $[-1, 0]$:

	z_i	$f[z_i]$	1st dd.	2nd dd.	3rd dd.
z_0	-1	0.0385	0.0740	0.8876	-1.8491
z_1	-1	0.0385	0.9615	-0.9615	0.0000
z_2	0	1.0000	0.0000	0.0000	0.0000
z_3	0	1.0000	0.0000	0.0000	0.0000

$$\begin{aligned} P_3(x) &= 0.038461538461538464 + 0.07396449704142012(x+1) \\ &\quad + 0.8875739644970414(x+1)^2 - 1.849112426035503(x+1)^2(x) \end{aligned}$$

On $[0, 1]$:

	z_i	$f[z_i]$	1st dd.	2nd dd.	3rd dd.
z_0	0	1.0000	0.0000	-0.9615	1.8491
z_1	0	1.0000	-0.9615	0.8876	0.0000
z_2	1	0.0385	-0.0740	0.0000	0.0000
z_3	1	0.0385	0.0000	0.0000	0.0000

$$\begin{aligned} P_3(x) &= 1.0 - 0.9615384615384616(x)^2 \\ &\quad + 1.849112426035503(x)^2(x-1) \end{aligned}$$

Question 1.4

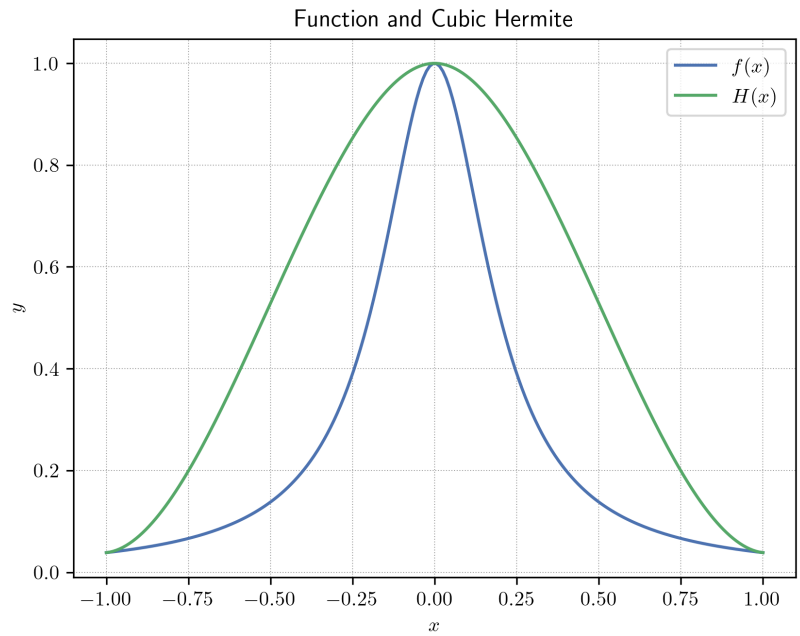
Plot $f(x)$, $H(x)$ and $P(x)$ on $x \in [-1, 1]$.

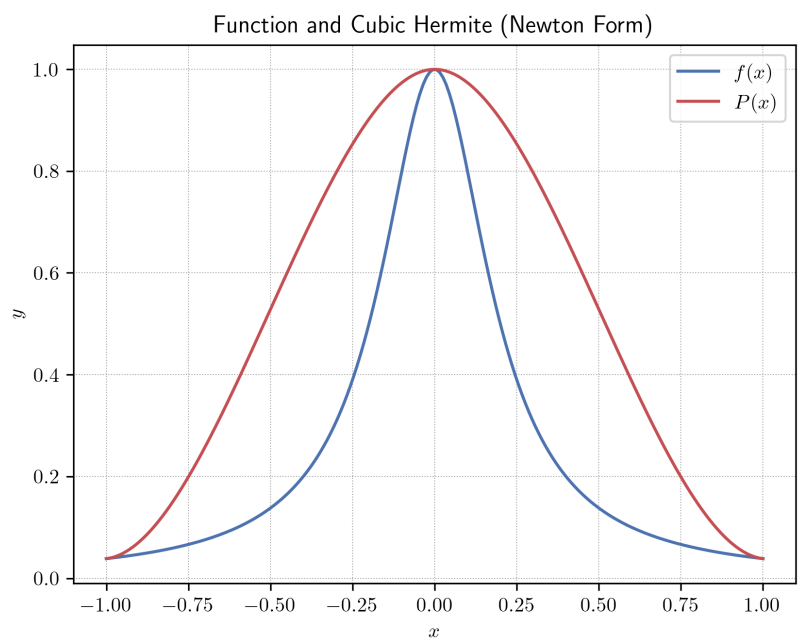
Solution. The following code was used to plot the functions:

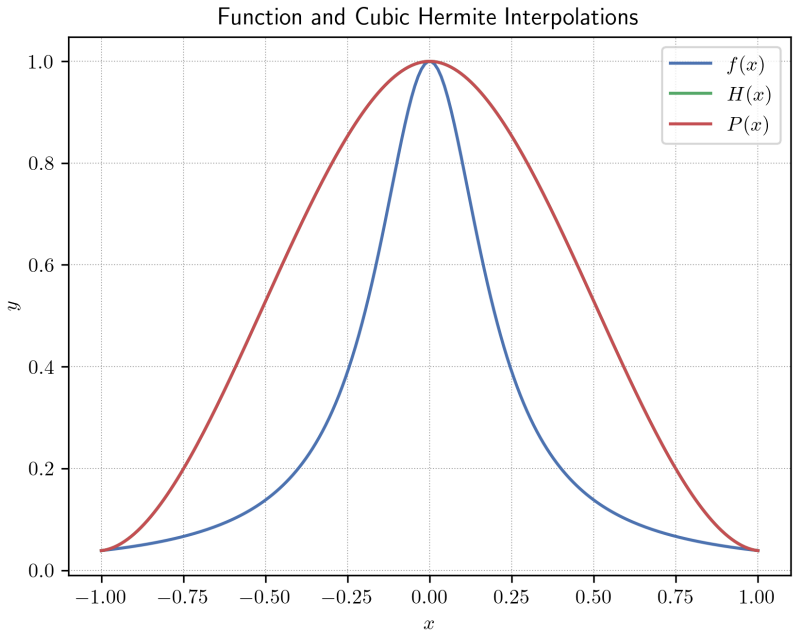
Listing 2: 1.4 Python

```python
import numpy as np
import matplotlib.pyplot as plt

if __name__ == "__main__":
    #<Some matplotlib styling and enable LaTeX>
    x = np.linspace(-1, 1, 200)
    f_y = f(x)

    conditions = [x <= 0, x > 0]
    h_y = np.piecewise(x, conditions, [h_1, h_2])
    p_y = np.concatenate(
        (calculate_hermite(nodes_1, x[:100]), calculate_hermite(nodes_2, x[100:]))
    )

    plt.plot(x, f_y, label="$f(x)$")
    plt.plot(x, h_y, label="$H(x)$")
    plt.xlabel("$x$")
    plt.ylabel("$y$")
    plt.legend()
    plt.title("Function and Cubic Hermite")
    plt.savefig("./outputs_3/hermite_plot_h.png", dpi=300)
    plt.cla()

    # Similar Plotting for P(x) and for all three together
```

The following plots were generated:







We observe that both Hermite interpolants $H(x)$ and $P(x)$ are identical and loosely follow the behavior of $f(x)$ on the interval $[-1, 1]$.

2 - Simple Analysis on Finite Difference

Consider the forward difference formula for the first derivative of a smooth function $f(x)$:

$$D_h f(x) = \frac{f(x+h) - f(x)}{h}$$

The total error for this approximation can be expressed as:

$$E(h) = \frac{C_1}{h} + C_2 h$$

with the optimal step size h:

$$h_{opt} = \sqrt{\frac{C_1}{C_2}}$$

where C_1 and C_2 are constants depending on $f(x)$, machine precision ε, and its derivatives.

Let $f(x) = e^x$, and take $x = 1.5$.

Question 2.1

Compute $f'(1.5)$ using the forward difference formula for a range of step sizes $h = 10^{-k}$, where $k = 1, 2, \ldots, 10$.

Solution. The following code was used to compute the forward difference approximations:

Listing 3: 2.1 Python

```python
import numpy as np

def forward_diff(f, x, step=1):
    return (f(x + step) - f(x)) / step

if __name__ == "__main__":
    with open("./outputs_3/forward_diff.txt", "w") as file:
        for k in range(1,11):
            h =10**(-k)
            f_prime = forward_diff(np.exp, 1.5, h)
            file.write(f'\\[D_h^{{({k})}}f(1.5) = {f_prime}\\]\n')
```

The following results were obtained for $D_h^{(k)} f(1.5)$:

$$D_h^{(1)} f(1.5) = 4.713433540570504$$

$$D_h^{(2)} f(1.5) = 4.5041723976187775$$

$$D_h^{(3)} f(1.5) = 4.483930662008362$$

$$D_h^{(4)} f(1.5) = 4.481913162264206$$

$$D_h^{(5)} f(1.5) = 4.4817114789097445$$

$$D_h^{(6)} f(1.5) = 4.48169131139764$$

$$D_h^{(7)} f(1.5) = 4.481689304114411$$

$$D_h^{(8)} f(1.5) = 4.481689064306238$$

$$D_h^{(9)} f(1.5) = 4.481689686031132$$

$$D_h^{(10)} f(1.5) = 4.48169501510165$$

Question 2.2

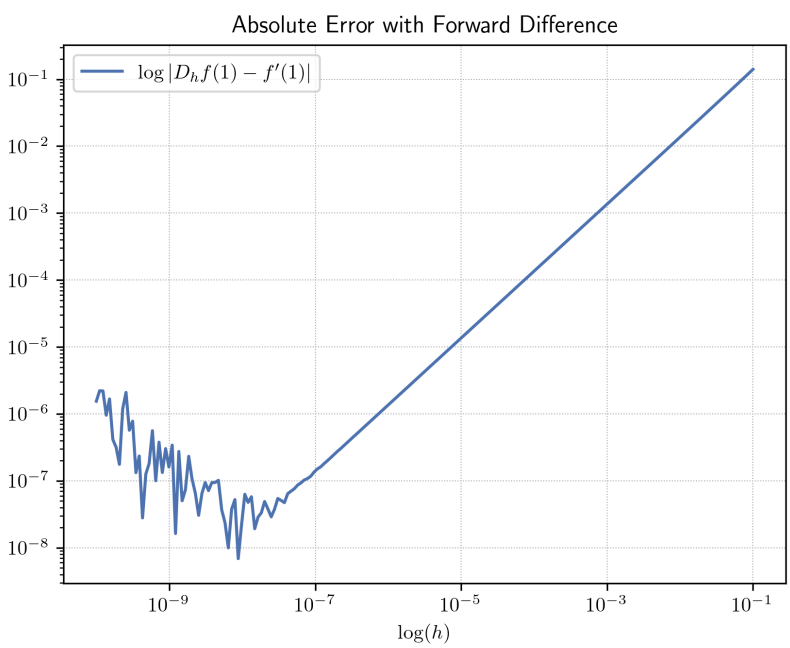
Plot the absolute error $|D_h f(1) - f'(1)|$ verses h on a log-log scale. (Report code)

Solution. The following code was used to compute and plot the absolute error:

Listing 4: 2.2 Python

```python
import numpy as np
import matplotlib.pyplot as plt

def calc_absolute_error(f, step, eval_x=1, diff_method=forward_diff):
    approx = diff_method(f, eval_x, step)
    return np.abs(approx - f(eval_x))

if __name__ == "__main__":
    # <Some matplotlib styling and enable LaTeX>
    h = np.logspace(-1, -10, 200)
    error = calc_absolute_error(np.exp, h)
    plt.loglog(h, error, label='$\\log|D_hf(1) - f\'(1)|$')
    plt.xlabel('$\\log(h)$')
    plt.title('Absolute Error in Forward Difference Approximation')
    plt.legend()
    plt.savefig('./outputs_3/forward_diff_graph.png', dpi=300)
    plt.show()
```

The resulting plot was:



We observe that machine error starts to occur as h gets smaller than approximately than 10^{-7}.

Question 2.3

Identify the h that minimizes the total error and compare it with the predicted h_{opt}.

Solution. From the plot above, we observe that the minimum error occurs around $h \approx 10^{-8}$.

Machine error is $\varepsilon \sim 10^{-16}$ for double precision floating point numbers. From the notes, we have that $C_1 \sim \varepsilon|f(1)|$ and $C_2 \sim \frac{|f''(1)|}{2}$. This resulted in $h_{opt} \propto \sqrt{\varepsilon} \approx 10^{-8}$, which aligns with our observed minimum error point.

Question 2.4

Use the one-sided three points scheme to approximate $f'(1.5)$, and repeat question 2.2.

Solution. The one-sided three-point finite difference formula for the first derivative is given by:

$$D_h f(x) = \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h}$$

The following code was used to compute and plot the absolute error using the three-point forward difference scheme:

Listing 5: 2.4 Python

```python
import numpy as np
import matplotlib.pyplot as plt

def three_point_forward_diff(f, x, step=1):
    return (-3 * f(x) + 4 * f(x + step) - f(x + 2 * step)) / (2 * step)

if __name__ == "__main__":
    with open("./outputs_3/three_point_forward_diff.txt", "w") as file:
        for k in range(1,11):
            h =10**(-k)
            f_prime = three_point_forward_diff(np.exp, 1.5, h)
            file.write(f'\\[D_h^{{({k})}}f(1.5) = {f_prime}\\]\n')

    # <Some matplotlib styling and enable LaTeX>
    h = np.logspace(-1, -10, 200)
    error = calc_absolute_error(np.exp, h, diff_method=three_point_forward_diff
    )

    plt.loglog(h, error, label='$\\log|D_hf(1) - f\'(1)|$')
    plt.xlabel('$\\log(h)$')
    plt.title('Absolute Error with Three Point Forward Difference')
    plt.legend()
    plt.savefig('./outputs_3/three_point_forward_diff_graph.png', dpi=300)
    plt.show()
```

Our approximation results for $D_h^{(k)} f(1.5)$ are:

$$D_h^{(1)} f(1.5) = 4.465575474195327$$

$$D_h^{(2)} f(1.5) = 4.4815385550327935$$

$$D_h^{(3)} f(1.5) = 4.481687575320414$$

$$D_h^{(4)} f(1.5) = 4.481689055397808$$

$$D_h^{(5)} f(1.5) = 4.481689070257033$$

$$D_h^{(6)} f(1.5) = 4.481689070079398$$

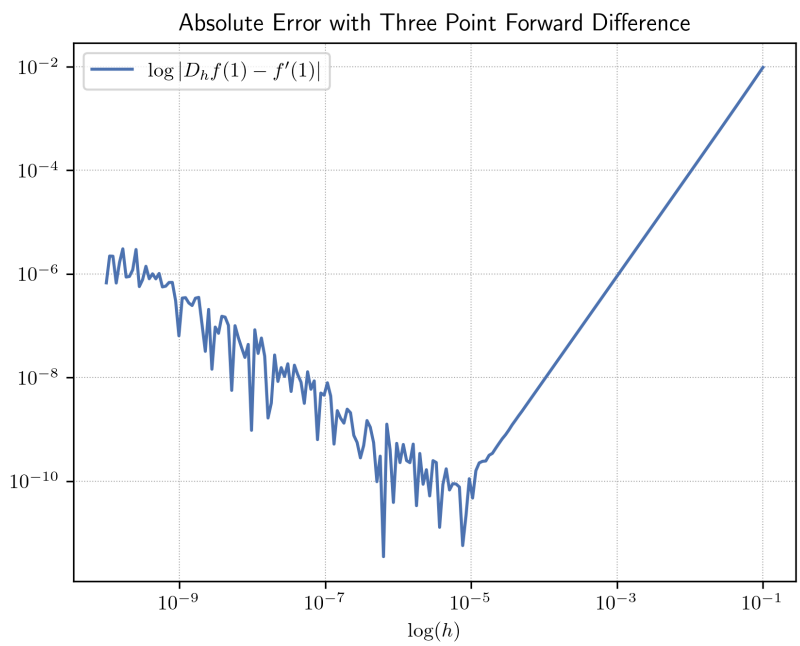
$$D_h^{(7)} f(1.5) = 4.481689086510698$$

$$D_h^{(8)} f(1.5) = 4.481688931079475$$

$$D_h^{(9)} f(1.5) = 4.481689686031132$$

$$D_h^{(10)} f(1.5) = 4.48169501510165$$

Also, the resulting plot was:



We observe that machine error starts to occur as h gets smaller than approximately than 10^{-5}.

3 - Solve Burger's Equation by Finite Difference Scheme

Consider the 1D viscous Burgers' equation:

$$u_t + uu_x = vu_{xx}, \quad x \in [0,1],\ t > 0,\ v = 0.2$$

Design, analyze, and implement a numerical method that is stable and achieves at least second-order accuracy in both space and time.

Question 3.1

Report finite difference scheme used.

Solution. We will be using RK4 with 2nd order finite differences for space.

First let us discretize our system. Let $x_0, x_1, ..., x_n$ be a partition of $[0,1]$ with step size $\Delta x = h = \frac{1}{n}$. Let $t_0, t_1, ..., t_m$ be a partition of $[0,T]$ with step size Δt. Let u_i be the numerical approximation of $u(x_i, t)$.

We can construct our schemes for u_x, u_{xx} from the Taylor expansion of u centered at x_i:

$$u(x) = u(x_i) + u_x(x_i)(x - x_i) + \frac{u_{xx}(x_i)}{2}(x - x_i)^2 + \frac{u_{xxx}(x_i)}{6}(x - x_i)^3 + ...$$

Which gives us for $x = x_i - h$ and $x = x_i + h$:

$$u(x_i - h) = u(x_i) - hu_x(x_i) + \frac{h^2}{2}u_{xx}(x_i) - \frac{h^3}{6}u_{xxx}(x_i) + O(h^4)$$

$$u(x_i + h) = u(x_i) + hu_x(x_i) + \frac{h^2}{2}u_{xx}(x_i) + \frac{h^3}{6}u_{xxx}(x_i) + O(h^4)$$

Approximate u_x centrally by:

$$u(x_i + h) - u(x_i - h) = 2hu_x(x_i) + O(h^3)$$

Which gives us the second-order scheme:

$$u_x(x_i) \approx \frac{u(x_i + h) - u(x_i - h)}{2h} + O(h^2)$$

Approximate u_{xx} with the usual second-order central difference:

$$u(x_i + h) - 2u(x_i) + u(x_i - h) = h^2 u_{xx}(x_i) + O(h^4)$$

Which gives us the second-order scheme:

$$u_{xx}(x_i) \approx \frac{u(x_i + h) - 2u(x_i) + u(x_i - h)}{h^2} + O(h^2)$$

Rearranging Burger's equation for u_t, we have:

$$u_t = -uu_x + vu_{xx}$$

Substituting in our finite difference approximations, we obtain the ODE:

$$\frac{du_i}{dt} \approx -u_i \cdot \frac{u_{i+1} - u_{i-1}}{2h} + v \cdot \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + O(h^2)$$

Which is $O(h^2) = O(\Delta x^2)$ accurate in space. Denote the right-hand side above as $F(u_i)$.

More compactly, write the system as:

$$\frac{du}{dt} = F(u) + O(\Delta x^2)$$

Where $u = [u_0, u_1, ..., u_n]^T$.

We will now use RK4 to approximate u^{n+1}, which is given by:

$$u^{n+1} = u^n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Such that:

$$\begin{aligned} k_1 &= F(u^n) \\ k_2 &= F\left(u^n + \frac{\Delta t}{2}k_1\right) \\ k_3 &= F\left(u^n + \frac{\Delta t}{2}k_2\right) \\ k_4 &= F\left(u^n + \Delta t k_3\right) \end{aligned}$$

RK4 is $O(\Delta t^4)$ accurate in time, so our overall scheme is $O(\Delta t^4) + O(\Delta x^2)$. We also note that RK4 would have CFL conditions for stability:

$$\Delta t \leq \frac{C_1 \Delta x}{\max |u|}, \quad \Delta t \leq \frac{C_2 \Delta x^2}{v}$$

Where C_1, C_2 are constants depending on the scheme. In practice, we can choose Δt based on the more restrictive of the two conditions above and pick smaller C_1, C_2 to be safe.

Question 3.2

Use the initial condition as a sine wave $\sin(2\pi x)$. Implement the scheme and plot the numerical solution at $T = 1$ second.

Solution. We have Burger's equation:

$$u_t + uu_x = vu_{xx}, \quad x \in [0,1],\ t > 0,\ v = 0.2$$

With periodic initial condition:

$$u(x,0) = \sin(2\pi x)$$

This means that $u(0,t) = u(1,t)$ for all $t \geq 0$. Since our data is periodic, no extra scheme is needed for the boundaries and we can just treat u as a circular array.

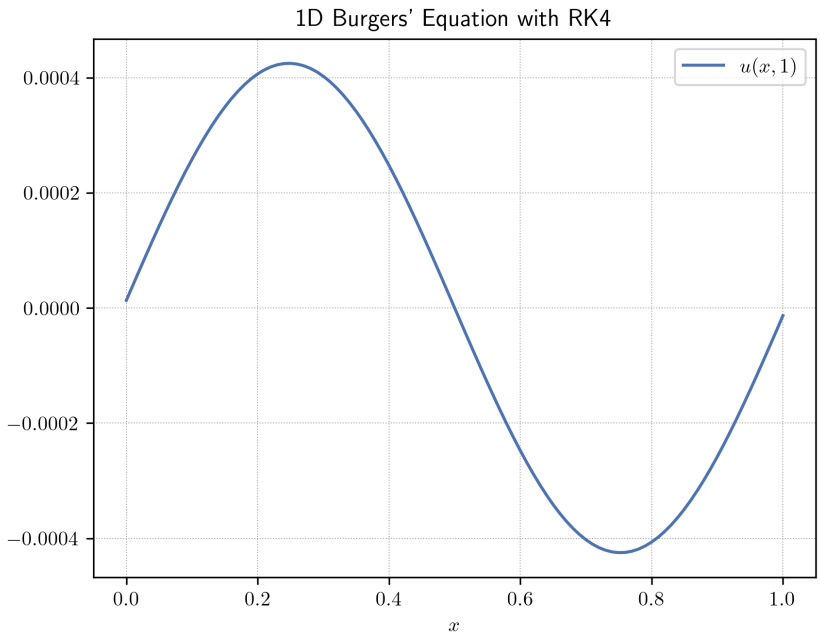
The following code was used to implement the scheme:

Listing 6: 3.2 Python

```python
import numpy as np
import matplotlib.pyplot as plt
from copy import deepcopy

def initial_condition(x):
    return np.sin(2 * np.pi * x)

def rk4_step(u, dt, dx, v):
    def F(u):
        # Compute spatial derivatives using central differences
        u_x = (np.roll(u, -1) - np.roll(u, 1)) / (2 * dx)
        u_xx = (np.roll(u, -1) - 2 * u + np.roll(u, 1)) / (dx**2)
        return -u * u_x + v * u_xx

    k1 = F(u)
    k2 = F(u + 0.5 * dt * k1)
    k3 = F(u + 0.5 * dt * k2)
    k4 = F(u + dt * k3)

    return u + (dt / 6) * (k1 + 2 * k2 + 2 * k3 + k4)

def burgers_rk4(dx, t_final, v=0.2, start_x=0, end_x=1, return_history=False):
    # Take ceil for num_nodes
    num_nodes = int((end_x - start_x) / dx) + 1
    x = np.linspace(start_x, end_x, num_nodes)

    # The CFL condition for diffusion is more restrictive
    # as for initial condition u = sin(2pi x), max|u| = 1
    # Pick C_2 = 0.5 to be safe
    dt = 0.5 * dx**2 / v
```

```python
    num_time_steps = int(np.ceil(t_final / dt))
    # Adjust dt to fit exactly into t_final
    dt = t_final / num_time_steps

    u = initial_condition(x)

    # Store history of u for Question 3.4
    if return_history:
        t = 0
        u_history = [deepcopy(u)]
        times = [t]

    for _ in range(num_time_steps):
        u = rk4_step(u, dt, dx, v)
        if return_history:
            t += dt
            u_history.append(deepcopy(u))
            times.append(t)

    if return_history:
        return x, u, np.array(times), np.array(u_history)

    return x, u

if __name__ == "__main__":
    dx = 1e-1
    t_final = 1
    x, u = burgers_rk4(dx, t_final)

    # <Some matplotlib styling and enable LaTeX>

    plt.plot(x, u, label=f'$u(x, {t_final})$')
    plt.title("1D Burgers' Equation with RK4")
    plt.xlabel("$x$")
    plt.legend()
    plt.savefig('./outputs_3/burgers_rk4.png', dpi=300)
    plt.show()
```

The resulting plot at $T = 1$ second was:



Question 3.3

Compute $||u - u_h||_{L^2}$ with a different refined mesh, and show your spatial convergence in a log-log plot.

Solution. The discrete L^2 norm was used to compute our error:

$$||u - u_h||_{L^2} = \sqrt{\Delta x \sum_{i=1}^{N} (u_i - u_{h,i})^2}$$

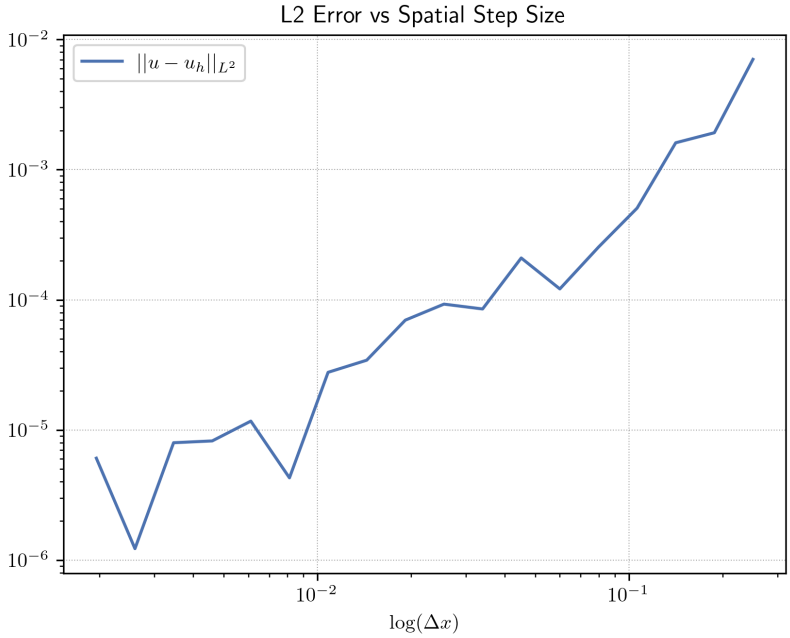
Where N is the number of spatial nodes in the coarser mesh, u_i is the exact solution at node i, and $u_{h,i}$ is the numerical solution at node i.

The following code was used to compute the L^2 error and plot spatial convergence:

Listing 7: 3.3 Python

```python
import numpy as np
import matplotlib.pyplot as plt

def discrete_l2(u_num, u_ref, dx):
    return np.sqrt(np.sum((u_num - u_ref)**2) * dx)

def spacial_convergence(dx_list, t_final, v=0.2):

    # dx_exact will serve as our "exact" solution
    # Use a finer dx than minimum dx in dx_list
    dx_exact = min(dx_list) / 4
    x_exact, u_exact = burgers_rk4(dx_exact, t_final, v)

    errors = []
    for dx in dx_list:
        x_approx, u_approx = burgers_rk4(dx, t_final, v)
        # Interpolate exact solution to the current mesh of x_approx
        u_exact_on_x_approx = np.interp(x_approx, x_exact, u_exact)
        error = discrete_l2(u_approx, u_exact_on_x_approx, dx)
        errors.append(error)
    return dx_list, errors

if __name__ == "__main__":
    # <Some matplotlib styling and enable LaTeX>
    dx_list = np.logspace(-2, -9, 18, base=2)
    dx, error = spacial_convergence(dx_list, t_final)
    plt.loglog(dx, error, label='$||u - u_h||_{L^2}$')
    plt.xlabel('$\\log(\\Delta x)$')
    plt.title('L2 Error vs Spatial Step Size')
    plt.legend()
    plt.savefig('./outputs_3/burgers_rk4_error.png', dpi=300)
    plt.show()
```

The plot generated for spatial convergence at $T = 1$ was:



A few notes:

1. $\Delta x = \frac{1}{2}$ has been excluded from the plot as it had favorable error due to the coarse mesh aligning well with the sine wave at $T = 1$.
2. The fluctuations in the error can be explained by additional interpolation error when mapping the exact solution to the coarser mesh and/or point 1 above.

Overall, we observe second-order spatial convergence as Δx decreases.

Question 3.4

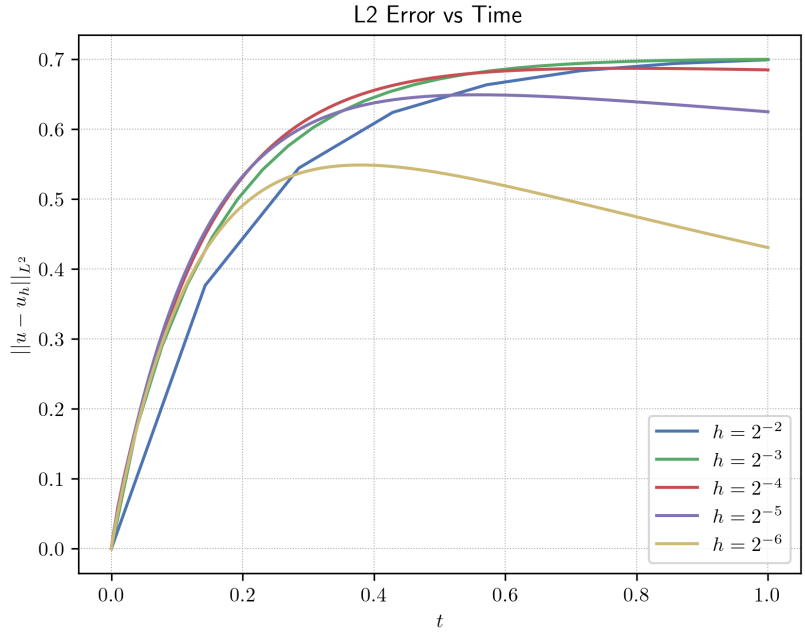
Plot your $||u - u_h||_{L^2}$ vs time.

Solution. The following code was used to compute and plot the L^2 error vs time:

Listing 8: 3.4 Python

```python
import numpy as np
import matplotlib.pyplot as plt

def norm_vs_time(dx, t_final, x_exact, u_exact_history, v=0.2):
    x_approx, _, times, u_history = burgers_rk4(dx, t_final, v, return_history=True)

    norms = []
    for u_approx, u_exact in zip(u_history, u_exact_history):
        u_exact_on_x_approx = np.interp(x_approx, x_exact, u_exact)
        norm = discrete_l2(u_approx, u_exact_on_x_approx, dx)
        norms.append(norm)
    return times, norms

if __name__ == "__main__":
    # <Some matplotlib styling and enable LaTeX>
    dx_list = np.logspace(-2, -6, 5, base=2)

    dx_exact = min(dx_list) / 4
    x_exact, _, _, u_exact_history = burgers_rk4(dx_exact, t_final, return_history=True)

    for index, dx in enumerate(dx_list):
        time, norms = norm_vs_time(dx, t_final, x_exact, u_exact_history)
        plt.plot(time, norms, label=f'$h = 2^{{{-index-2}}}$')
    plt.xlabel('$t$')
    plt.ylabel(r'$||u - u_h||_{L^2}$')
    plt.title('L2 Error vs Time')
    plt.legend()
    plt.savefig('./outputs_3/burgers_rk4_error_vs_time.png', dpi=300)
    plt.show()
```

The plot generated for L^2 error vs time was:



We can see that the L^2 error generally increases with time and then stabilizes. The increase makes sense as numerical errors accumulate over time steps.

Finer meshes (smaller h) consistently yield lower errors throughout the simulation at $T = 1$.

Appendix

The complete code used for this assignment is provided in the appendix for reference. Files can be accessed directly at this GitHub repository.

Listing 9: question_1.py

```
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return 1 / (1 + 25 * x**2)


def f_prime(x):
    return -50 * x / (1 + 25 * x**2) ** 2


def h_1(x):
    return (-625 / 338) * x**3 - (475 / 169) * x**2 + 1


def h_2(x):
    return (625 / 338) * x**3 - (475 / 169) * x**2 + 1


def hermite_coefficients(nodes, f=f, f_prime=f_prime):
    # Parameterize nodes for Hermite interpolation
    z = np.concatenate((nodes, nodes))
    sorted_indexes = z.argsort()
    z = z[sorted_indexes]
    num_nodes = len(z)

    # Set up dd table with zeroth dd
    dd_table = np.array([[f(zi) for zi in z]])

    # First divided difference
    f_prime_nodes = np.array([f_prime(xi) for xi in nodes])
    zeros = np.zeros(len(nodes))
    first_dd = np.concatenate((f_prime_nodes, zeros))[sorted_indexes]
    for j in range(num_nodes - 1):
        if j % 2 == 1:
            first_dd[j] = (dd_table[0, j + 1] - dd_table[0, j]) / (z[j + 1] - z[j])
    dd_table = np.vstack([dd_table, first_dd])

    # Remaining Divided Differences
    for i in range(2, num_nodes):
```

```
        ith_dd = np.zeros(num_nodes)

        for j in range(num_nodes - i):
            # Calculate ith divided differences
            ith_dd[j] = (dd_table[i - 1, j + 1] - dd_table[i - 1, j]) / (
                z[j + i] - z[j]
            )

        dd_table = np.vstack([dd_table, ith_dd])

    coefficients = np.array([dd_table[i, 0] for i in range(dd_table.shape[0])])
    return coefficients, dd_table.T, z


def generate_hermite(nodes, n):
    # Get coefficients
    a = hermite_coefficients(nodes)[0]
    # Start with function and constant term
    equation = f"P_{{{2 * n + 1}}}(x) = {a[0]}"
    w = []
    # Build (x - xi) terms
    for xi in nodes:
        if abs(xi) <= 1e-14:
            w.append("(x)")
        elif xi < 0:
            w.append(f"(x + {abs(xi)})")
        else:
            w.append(f"(x - {xi})")
    b = ""

    # Build polynomial string
    for i in range(1, len(a)):
        for j in range(i):
            # Multiply (x - xi) terms
            b += "^2" if j % 2 == 1 else w[int(j / 2)]
        # Multiply (x - xi) product with current coefficient and add term
        if a[i] > 0:
            equation += f" + {a[i]}{b}"
        elif a[i] < 0:
            equation += f" - {abs(a[i])}{b}"
        b = ""

    return equation


def calculate_hermite(nodes, x_coords=[], function=f):
    # Get coefficients
```

```python
    a, z = hermite_coefficients(nodes, function)[:3:2]
    # If no x_coords provided, use nodes as x_coords
    if len(x_coords) == 0:
        x_coords = nodes
    # Start with constant term
    y = a[0]
    # Build (x - zi) terms
    w = [(x_coords - zi) for zi in z]
    # Temporary variable to hold (x - zi) product
    b = 1
    for i in range(1, len(a)):
        for j in range(i):
            # Multiply (x - zi) terms
            b *= w[j]
        # Multiply (x - zi) product with current coefficient
        y += a[i] * b
        b = 1

    return y


if __name__ == "__main__":
    nodes_1 = np.array([-1, 0])
    equation_1 = generate_hermite(nodes_1, 1)
    table_1, z_1 = hermite_coefficients(nodes_1)[1:3]

    nodes_2 = np.array([0, 1])
    equation_2 = generate_hermite(nodes_2, 1)
    table_2, z_2 = hermite_coefficients(nodes_2)[1:3]

    equations = [equation_1, equation_2]
    tables = [table_1, table_2]
    z = [z_1, z_2]

    with open("./outputs_3/hermite.txt", "w") as file:
        for k in range(2):
            file.write(equations[k] + "\n\n")

            file.write("\\begin{center}\n")
            file.write("\\begin{tabular}{|c|c|c|c|c|c|}\n")
            file.write("\\hline\n")
            file.write("& $z_i$ & $f[z_i]$ & 1st dd. & 2nd dd. & 3rd dd. \\\\\n")
            file.write("\\hline\n")
            for i in range(tables[k].shape[0]):
                file.write(f"$z_{i}$ & ${z[k][i]}$ ")
                for j in range(tables[k].shape[1]):
```

```
                    file.write(f"& ${tables[k][i, j]:.4f}$ ")
                file.write("\\\\ \n")
            file.write("\\hline\n")
            file.write("\\end{tabular}\n")
            file.write("\\end{center}\n\n")

    plt.rcParams["text.usetex"] = True
    plt.rcParams["axes.grid"] = True
    plt.rc("grid", color="#a6a6a6", linestyle="dotted", linewidth=0.5)
    plt.style.use("seaborn-v0_8-deep")
    prop_cycle = plt.rcParams["axes.prop_cycle"]
    default_colors = prop_cycle.by_key()["color"]

    x = np.linspace(-1, 1, 200)
    f_y = f(x)

    conditions = [x <= 0, x > 0]
    h_y = np.piecewise(x, conditions, [h_1, h_2])
    p_y = np.concatenate(
        (calculate_hermite(nodes_1, x[:100]), calculate_hermite(nodes_2, x[100:]))
    )

    plt.plot(x, f_y, label="$f(x)$")
    plt.plot(x, h_y, label="$H(x)$")
    plt.xlabel("$x$")
    plt.ylabel("$y$")
    plt.legend()
    plt.title("Function and Cubic Hermite")
    plt.savefig("./outputs_3/hermite_plot_h.png", dpi=300)
    plt.cla()

    plt.plot(x, f_y, label="$f(x)$")
    plt.plot(x, p_y, label="$P(x)$", color = default_colors[2])
    plt.xlabel("$x$")
    plt.ylabel("$y$")
    plt.legend()
    plt.title("Function and Cubic Hermite (Newton Form)")
    plt.savefig("./outputs_3/hermite_plot_p.png", dpi=300)
    plt.cla()

    plt.plot(x, f_y, label="$f(x)$")
    plt.plot(x, h_y, label="$H(x)$")
    plt.plot(x, p_y, label="$P(x)$")
    plt.xlabel("$x$")
    plt.ylabel("$y$")
    plt.legend()
```

```
    plt.title("Function and Cubic Hermite Interpolations")
    plt.savefig("./outputs_3/hermite_plot.png", dpi=300)
```

Listing 10: question_2.py

```
import numpy as np
import matplotlib.pyplot as plt

def forward_diff(f, x, step=1):
    return (f(x + step) - f(x)) / step

def three_point_forward_diff(f, x, step=1):
    return (-3 * f(x) + 4 * f(x + step) - f(x + 2 * step)) / (2 * step)

def calc_absolute_error(f, step, eval_x=1, diff_method=forward_diff):
    approx = diff_method(f, eval_x, step)
    return np.abs(approx - f(eval_x))

if __name__ == "__main__":
    with open("./outputs_3/forward_diff.txt", "w") as file:
        for k in range(1,11):
            h =10**(-k)
            f_prime = forward_diff(np.exp, 1.5, h)
            file.write(f'\\[D_h^{{({k})}}f(1.5) = {f_prime}\\]\n')

    plt.rcParams["text.usetex"] = True
    plt.rcParams["axes.grid"] = True
    plt.rc("grid", color="#a6a6a6", linestyle="dotted", linewidth=0.5)
    plt.style.use("seaborn-v0_8-deep")

    h = np.logspace(-1, -10, 200)
    error = calc_absolute_error(np.exp, h)

    plt.loglog(h, error, label='$\\log|D_hf(1) - f\'(1)|$')
    plt.xlabel('$\\log(h)$')
    plt.title('Absolute Error with Forward Difference')
    plt.legend()
    plt.savefig('./outputs_3/forward_diff_graph.png', dpi=300)
    plt.show()

    with open("./outputs_3/three_point_forward_diff.txt", "w") as file:
        for k in range(1,11):
            h =10**(-k)
            f_prime = three_point_forward_diff(np.exp, 1.5, h)
            file.write(f'\\[D_h^{{({k})}}f(1.5) = {f_prime}\\]\n')

    h = np.logspace(-1, -10, 200)
    error = calc_absolute_error(np.exp, h, diff_method=three_point_forward_diff
```

```
    )

    plt.loglog(h, error, label='$\\log|D_hf(1) - f\'(1)|$')
    plt.xlabel('$\\log(h)$')
    plt.title('Absolute Error with Three Point Forward Difference')
    plt.legend()
    plt.savefig('./outputs_3/three_point_forward_diff_graph.png', dpi=300)
    plt.show()
```

Listing 11: question_3.py

```
import numpy as np
import matplotlib.pyplot as plt
from copy import deepcopy

def initial_condition(x):
    return np.sin(2 * np.pi * x)

def rk4_step(u, dt, dx, v):
    def F(u):
        # Compute spatial derivatives using central differences
        u_x = (np.roll(u, -1) - np.roll(u, 1)) / (2 * dx)
        u_xx = (np.roll(u, -1) - 2 * u + np.roll(u, 1)) / (dx**2)
        return -u * u_x + v * u_xx

    k1 = F(u)
    k2 = F(u + 0.5 * dt * k1)
    k3 = F(u + 0.5 * dt * k2)
    k4 = F(u + dt * k3)

    return u + (dt / 6) * (k1 + 2 * k2 + 2 * k3 + k4)

def burgers_rk4(dx, t_final, v=0.2, start_x=0, end_x=1, return_history=False):
    # Take ceil for num_nodes
    num_nodes = int((end_x - start_x) / dx) + 1
    x = np.linspace(start_x, end_x, num_nodes)

    # The CFL condition for diffusion is more restrictive
    # as for initial condition u = sin(2pi x), max|u| = 1
    # Pick C_2 = 0.5 to be safe
    dt = 0.5 * dx**2 / v
    num_time_steps = int(np.ceil(t_final / dt))
    # Adjust dt to fit exactly into t_final
    dt = t_final / num_time_steps

    u = initial_condition(x)

    # Store history of u for Question 3.4
```

```
    if return_history:
        t = 0
        u_history = [deepcopy(u)]
        times = [t]

    for _ in range(num_time_steps):
        u = rk4_step(u, dt, dx, v)
        if return_history:
            t += dt
            u_history.append(deepcopy(u))
            times.append(t)

    if return_history:
        return x, u, np.array(times), np.array(u_history)

    return x, u

def discrete_l2(u_num, u_ref, dx):
    return np.sqrt(np.sum((u_num - u_ref)**2) * dx)

def spacial_convergence(dx_list, t_final, v=0.2):

    # dx_exact will serve as our "exact" solution
    # Use a finer dx than minimum dx in dx_list
    dx_exact = min(dx_list) / 4
    x_exact, u_exact = burgers_rk4(dx_exact, t_final, v)

    errors = []
    for dx in dx_list:
        x_approx, u_approx = burgers_rk4(dx, t_final, v)
        # Interpolate exact solution to the current mesh of x_approx
        u_exact_on_x_approx = np.interp(x_approx, x_exact, u_exact)
        error = discrete_l2(u_approx, u_exact_on_x_approx, dx)
        errors.append(error)
    return dx_list, errors

def norm_vs_time(dx, t_final, x_exact, u_exact_history, v=0.2):
    x_approx, _, times, u_history = burgers_rk4(dx, t_final, v, return_history=
    True)

    norms = []
    for u_approx, u_exact in zip(u_history, u_exact_history):
        u_exact_on_x_approx = np.interp(x_approx, x_exact, u_exact)
        norm = discrete_l2(u_approx, u_exact_on_x_approx, dx)
        norms.append(norm)
    return times, norms

```

```python
if __name__ == "__main__":
    dx = 1e-2
    t_final = 1
    x, u = burgers_rk4(dx, t_final)

    plt.rcParams["text.usetex"] = True
    plt.rcParams["axes.grid"] = True
    plt.rc("grid", color="#a6a6a6", linestyle="dotted", linewidth=0.5)
    plt.style.use("seaborn-v0_8-deep")

    plt.plot(x, u, label=f'$u(x, {t_final})$')
    plt.title("1D Burgers' Equation with RK4")
    plt.xlabel("$x$")
    plt.legend()
    plt.savefig('./outputs_3/burgers_rk4.png', dpi=300)
    plt.cla()

    dx_list = np.logspace(-2, -9, 18, base=2)
    dx, error = spacial_convergence(dx_list, t_final)
    plt.loglog(dx, error, label='$||u - u_h||_{L^2}$')
    plt.xlabel('$\\log(\\Delta x)$')
    plt.title('L2 Error vs Spatial Step Size')
    plt.legend()
    plt.savefig('./outputs_3/burgers_rk4_error.png', dpi=300)
    plt.show()

    dx_list = np.logspace(-2, -6, 5, base=2)

    dx_exact = min(dx_list) / 4
    x_exact, _, _, u_exact_history = burgers_rk4(dx_exact, t_final,
    return_history=True)

    for index, dx in enumerate(dx_list):
        time, norms = norm_vs_time(dx, t_final, x_exact, u_exact_history)
        plt.plot(time, norms, label=f'$h = 2^{{{-index-2}}}$')
    plt.xlabel('$t$')
    plt.ylabel(r'$||u - u_h||_{L^2}$')
    plt.title('L2 Error vs Time')
    plt.legend()
    plt.savefig('./outputs_3/burgers_rk4_error_vs_time.png', dpi=300)
    plt.show()
```